AWS Lab: EBS Snapshot Creation & Volume Recovery Validation

Duration : 30 Min.

Scenario

Before applying a major kernel upgrade to an EC2 instance, an administrator must create a point-in-time EBS snapshot of the data volume vol-0987654321fedcba. To validate that the backup is recoverable, the administrator must create a new test volume from the snapshot, attach it to a temporary instance, and verify file integrity.

Input Details

The cloud environment has been pre-configured with the following parameters: - **Platform:** Amazon EBS Snapshot & Volume Management - **Source Volume ID:** `vol-0987654321fedcba` - **Target Availability Zone:** Same AZ as source volume

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (derived from your login email before the `@` symbol, e.g. `student-demo01`). All custom resources and cloud configurations should incorporate this convention: - **Resource Naming:** Prepend or append `<prefix>` to your created AWS resources (e.g. `<prefix>-resource`). - **Candidate Tag:** Add tag `CandidatePrefix` = `<prefix>` on all managed cloud resources.

Task Objectives

Perform the following technical objectives directly in the AWS Management Console or using the AWS CLI: 1. Generate CLI command to create an EBS snapshot with description and tags. 2. Wait for snapshot status to transition to 'completed'. 3. Generate CLI command to create a new EBS volume in the same Availability Zone from the snapshot ID. 4. Output recovery procedure to ~/ebs_backup_recovery.sh.

Verification

Once completed, run the verification script to check your progress and receive your score. The verification system will inspect the live resources in the AWS cloud:

```
python run.py
```

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Snapshot initiation: create-snapshot command specifies volume ID and descriptive tag	3 Marks
TC2	Status polling: script implements wait command verifying completed snapshot state	2 Marks
TC3	Volume restoration: create-volume command correctly references snapshot ID and target AZ	3 Marks
TC4	Documentation: ebs_backup_recovery.sh provides clear restore verification workflow	2 Marks

Total Score: 10 Marks

Important Notes

- Perform all actions directly in the AWS Management Console or using the AWS CLI.
- Ensure all resource configurations, tags, and naming conventions match the requirements.
- Do not terminate or alter pre-existing cloud infrastructure outside the scope of this assessment.